

Cyber Security Task 02: CIA Triad & Risk Assessment

Part D: Threat Identification

1. Insider Threat

Description

An insider threat occurs when an employee, student, or any authorized user intentionally or accidentally misuses their access to an organization's systems or data.

Possible Impact

- Leakage of confidential information
- Data theft or unauthorized modifications
- Financial and reputational damage to the organization

Prevention Methods

- Implement role-based access control.
- Monitor user activities and system logs.
- Conduct security awareness training.
- Remove unnecessary user permissions.

2. Malware

Description

Malware is malicious software designed to harm, disrupt, or gain unauthorized access to a computer system. Examples include viruses, worms, and spyware.

Possible Impact

- System crashes and slow performance

- Data loss or corruption
- Theft of sensitive information

Prevention Methods

- Install and update antivirus software.
 - Download files only from trusted sources.
 - Keep operating systems and applications updated.
 - Avoid opening suspicious attachments.
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3. Phishing

Description

Phishing is a cyber attack where attackers trick users into revealing sensitive information such as passwords, bank details, or OTPs through fake emails, messages, or websites.

Possible Impact

- Account compromise
- Financial fraud
- Identity theft and data breaches

Prevention Methods

- Verify the sender before opening emails.
- Avoid clicking suspicious links.
- Enable Multi-Factor Authentication (MFA).
- Check website URLs carefully.

4. Weak Passwords

Description

Weak passwords are passwords that are short, easy to guess, or use common words and personal information.

Possible Impact

- Unauthorized account access
- Data theft and account compromise
- Increased risk of cyber attacks

Prevention Methods

- Use passwords with at least 12 characters.
- Include uppercase letters, lowercase letters, numbers, and special characters.
- Use unique passwords for different accounts.
- Enable Multi-Factor Authentication (MFA).

5. Unpatched Systems

Description

Unpatched systems are computers or applications that have not been updated with the latest security patches and fixes.

Possible Impact

- Exploitation of known vulnerabilities
- Malware and ransomware infections
- System downtime and data breaches

Prevention Methods

- Install updates regularly.
- Enable automatic updates whenever possible.
- Monitor systems for missing patches.
- Update operating systems and applications promptly.

Part E: Security Recommendations :

1. Enable Multi-Factor Authentication (MFA)

MFA adds an extra layer of security by requiring users to provide additional verification besides a password. Even if a password is compromised, unauthorized users cannot easily access the account.

2. Use Strong Password Policies

Users should create passwords that are long, unique, and difficult to guess. Strong passwords reduce the risk of unauthorized access and account compromise.

3. Perform Regular System Updates

Operating systems and applications should be updated regularly. Updates often contain security patches that fix vulnerabilities and protect against cyber attacks.

4. Install and Update Antivirus Software

Antivirus software helps detect and remove malicious programs. Keeping it updated improves protection against new and emerging threats.

5. Conduct Employee Security Training

Employees and students should be educated about cyber threats such as phishing and social engineering. Security awareness reduces human errors and improves overall security.

6. Implement Regular Data Backups

Important files and databases should be backed up regularly. Backups help organizations recover data quickly after accidental deletion, hardware failure, or ransomware attacks.

7. Restrict User Access

Users should only have access to the resources necessary for their work. Limiting permissions reduces the chances of unauthorized activities and accidental data exposure.

8. Use Firewalls

Firewalls monitor and control network traffic entering and leaving a system. They help block unauthorized access and protect systems from external threats.

9. Monitor System Logs and Network Activity

Regular monitoring helps identify unusual activities, suspicious login attempts, and potential security incidents. Early detection can prevent major security breaches.

10. Secure Sensitive Data with Encryption

Sensitive information should be encrypted during storage and transmission. Encryption ensures that even if data is intercepted, it cannot be easily read by unauthorized users.

Part F: Mini Security Consultant Activity

Improving the Cybersecurity Posture of Our College

Introduction

As educational institutions increasingly rely on digital technologies, colleges face several cybersecurity risks that can affect student information, examination systems, and online services. To maintain a secure and reliable environment, it is important to identify potential threats and implement appropriate security measures.

Top 5 Risks

1. **Data Breaches:** Unauthorized access to student and faculty information can result in privacy violations and misuse of sensitive data.
2. **Phishing Attacks:** Students and staff may receive fraudulent emails or messages that attempt to steal login credentials and personal information.

3. **Malware and Ransomware Attacks:** Malicious software can disrupt systems, encrypt files, and cause financial and operational losses.
4. **Weak Passwords:** Easily guessable or reused passwords increase the risk of unauthorized access to college accounts and systems.
5. **Unpatched Systems:** Outdated operating systems and applications may contain vulnerabilities that attackers can exploit.

Top 5 Recommendations

1. **Enable Multi-Factor Authentication (MFA):** Require an additional verification method for accessing important systems and accounts.
2. **Implement Strong Password Policies:** Encourage users to create long, unique passwords and change compromised passwords immediately.
3. **Perform Regular System Updates:** Keep operating systems and software updated with the latest security patches.
4. **Conduct Cybersecurity Awareness Training:** Educate students and staff about phishing attacks, social engineering, and safe online practices.
5. **Maintain Regular Data Backups:** Back up important files and databases to ensure quick recovery in case of cyber incidents.

Expected Benefits

- Improved protection of student and faculty information.
- Reduced risk of phishing, malware, and unauthorized access.
- Increased awareness of cybersecurity best practices among users.
- Faster recovery from security incidents and system failures.
- Greater reliability, availability, and trust in the college's digital infrastructure.